

Task 2: Pandas & Dataset Handling

Introduction

Task 2 focused on learning the fundamentals of **Pandas**, one of the most widely used Python libraries for data manipulation and analysis. The main objective of this task was to understand how structured datasets can be loaded, organized, inspected, and analyzed using Python. A student performance dataset was used to demonstrate different Pandas operations and understand how data analysts work with real-world tabular data.

Creating and Loading Data

The task began with the creation of a **DataFrame** using Pandas. A DataFrame is a two-dimensional data structure consisting of rows and columns, similar to a spreadsheet or database table. This helped demonstrate how different types of information can be organized into a structured format. A student performance dataset was then stored in a **CSV (Comma-Separated Values)** file and loaded into Pandas using the `read_csv()` function. This demonstrated how external datasets can be imported into Python for further processing and analysis.

Exploring the Dataset

After loading the dataset, several Pandas functions were used to understand its structure. The `head()` function was used to display the first few records, while `tail()` was used to display the last few records. These functions are useful for quickly examining a dataset without displaying every record. The `shape` attribute was used to determine the number of rows and columns in the dataset. The `columns` attribute was used to identify the names of all available columns, such as Student ID, Name, Gender, Study Hours, Attendance, Mathematics, Science, and English. The `dtypes` attribute was also used to determine the data type of each column. The `info()` function provided a detailed overview of the dataset, including entries, non-null values, column types, and memory usage.

Statistical Analysis

The `describe()` function was used to obtain basic statistical information about the numerical columns. It provided values such as count, mean, standard deviation, minimum, maximum, and percentile values. This allowed the dataset to be understood from a statistical perspective and helped identify general patterns in student performance.

Selecting and Filtering Data

The task also demonstrated how specific columns can be selected from a DataFrame. For example, the Name, Mathematics, Science, and English columns were selected for focused analysis. Pandas filtering operations were then performed to identify students meeting specific conditions. For example, students who scored 90 or above in Mathematics were identified, and students with attendance of at least 90 percent were filtered. Multiple conditions were also combined to identify students who studied for a certain number of hours and maintained good attendance.

Learning Outcome

Through this task, the student gained a basic understanding of how Pandas can be used to handle structured datasets efficiently. The task demonstrated the complete initial process of loading, inspecting, understanding, selecting, and filtering data. These skills are essential for data analysis.

because properly understanding a dataset is an important step before performing data cleaning, visualization, or machine learning. Overall, Task 2 provided a strong foundation in **Pandas and dataset handling** and prepared the student for the next stages of the assignment, particularly data cleaning, visualization, and student performance analysis.